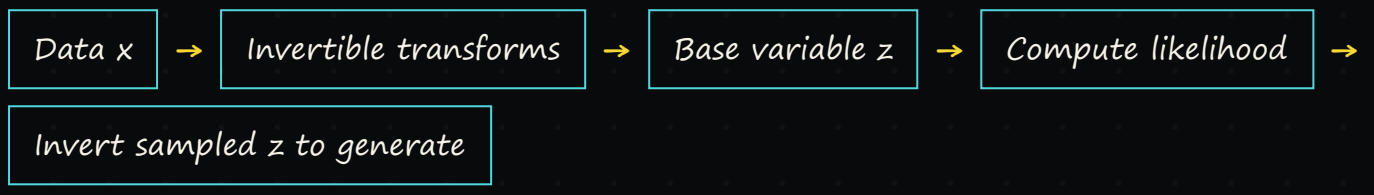
GENERATIVE DEEP LEARNING | CHAPTER 6

Normalizing Flow Models

Transform a simple base density with invertible functions so that both sampling and exact density evaluation remain computationally tractable.

Original study notes based on David Foster, Second edition, 2023. Book pp. 167–188. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.



Detailed Notes

Normalizing flows and change of variables

Source: book pp. 167–171 / PDF p. 195

Let z=f(x) map observations into a simple known base distribution. The probability density must change when the mapping expands or contracts volume. A normalizing flow tracks that correction exactly through a composition of invertible transforms.

$$\log p_X(x) = \log p_Z(f(x)) + \log\left|\det J_f(x)\right|$$

If a source defines the generator as x=g(z), the equivalent expression subtracts the generator's log determinant. State the mapping direction before choosing a sign. Unlike a VAE, a conventional flow does not reduce continuous dimension arbitrarily while retaining a simple bijection.

Jacobian determinant and composition

Source: book pp. 172–173 / PDF p. 200

The Jacobian records how each output changes with each input. Its absolute determinant measures local volume scaling. Large unrestricted Jacobians are expensive to evaluate, so useful flow architectures impose structure without discarding invertibility.

$$\log\left|\det J_{f_K \circ \dots \circ f_1}\right| = \Sigma_k \log\left|\det J_{f_k}\right|$$

A triangular Jacobian has determinant equal to the product of its diagonal entries, turning log determinants into sums. Zero determinant breaks invertibility. Numerical stability and finite scales matter just as much as the symbolic formula.

RealNVP and the two-moons dataset

Source: book pp. 174–180 / PDF p. 202

The two-moons project visualizes how a simple Gaussian can be warped into a curved multimodal distribution. An affine coupling layer splits variables, leaves one part unchanged, and scales/translates the other using functions of the unchanged part.

$$y_A = x_A$$
$$y_B = x_B \odot \exp(s(x_A)) + t(x_A)$$
$$\text{log determinant} = \Sigma\, s(x_A)$$

The s and t networks need not be invertible because their input remains available. Inversion subtracts t and multiplies by exp(−s). Alternate masks/permutations across layers so every coordinate can change.

Training, density analysis, and sampling

Source: book pp. 181–185 / PDF p. 209

Maximize the base log density plus accumulated log determinants for real observations. Choose and consistently implement whether the forward direction maps data to base or base to data. Generate by sampling the known base and applying the inverse data-normalizing map.

Inspect transformed data, base-distribution fit, learned density contours, new samples, and reconstruction through forward/inverse composition. Finite precision can affect round-trip accuracy. Exact likelihood does not automatically imply perceptual similarity or good out-of-distribution detection.

Glow, FFJORD, and architectural trade-offs

Source: book pp. 186–188 / PDF p. 214

Glow extends image flows with learned invertible channel mixing, including invertible 1×1 convolutions, avoiding only fixed permutations. **FFJORD** models a continuous flow using a neural ordinary differential equation and tracks density changes through divergence/trace calculations.

Flows give a strong connection between sampling and density evaluation but constrain architecture and can be costly at high dimension. Continuous flows replace discrete layers with an ODE solver, introducing tolerances, solver work, and numerical considerations. Their tractability should not be confused with zero computational cost.

Revision Check

1. Which direction is f in your density equation?
2. Why is the coupling-layer Jacobian triangular?
3. Why do coupling masks need to alternate?
4. How would you check invertibility numerically?

5. What changes with a continuous-time flow?

Invertibility alone is insufficient: a practical flow also needs an efficient, numerically stable log-determinant calculation.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.